

Differential effects of monounsaturated, polyunsaturated and saturated fat ingestion on glucose-stimulated insulin secretion, sensitivity and clear...

AIMS/HYPOTHESIS: Prolonged elevation of plasma specific fatty acids may exert differential effects on glucose-stimulated insulin secretion (GSIS), insulin sensitivity and clearance. **SUBJECTS AND METHODS:** We examined the effect of oral ingestion, at regular intervals for 24 h, of an emulsion containing either predominantly monounsaturated (MUFA), polyunsaturated (PUFA) or saturated (SFA) fat or water (control) on GSIS, insulin sensitivity and insulin clearance in seven overweight or obese, non-diabetic humans. Four studies were conducted in each individual in random order, 4-6 weeks apart. Twenty-four hours after initiation of oral ingestion, subjects underwent a 2 h, 20 mmol/l hyperglycaemic clamp to assess GSIS, insulin sensitivity and insulin clearance. **RESULTS:** Following oral ingestion of any of the three fat emulsions over 24 h, plasma NEFAs were elevated by approximately 1.5- to 2-fold over the basal level. Ingestion of any of the three fat emulsions resulted in reduction in insulin clearance, and SFA ingestion reduced insulin sensitivity. PUFA ingestion was associated with an absolute reduction in GSIS, whereas insulin secretion failed to compensate for insulin resistance in subjects who ingested SFA. **CONCLUSIONS/INTERPRETATION:** Oral ingestion of fats with differing degrees of saturation resulted in different effects on insulin secretion and action. PUFA ingestion resulted in an absolute reduction in insulin secretion and SFA ingestion induced insulin resistance. Failure of insulin secretion to compensate for insulin resistance implies impaired beta cell function in the SFA study.